

Multi-Regime Reinforcement Learning for Adaptive Portfolio Optimization

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Abstract—This paper presents a technical framework that integrates Hidden Markov Model (HMM)-based regime detection with a Reinforcement Learning (RL) agent for dynamic portfolio optimization. The approach aims to improve risk-adjusted returns by allowing the policy to adapt to changing market conditions. The model identifies latent market regimes such as bull, bear, and neutral markets using volatility and correlation structures, and then applies Proximal Policy Optimization (PPO) to learn allocation strategies across multiple assets. Experimental evaluation demonstrates that this regime-aware RL framework achieves superior Sharpe ratios and lower drawdowns compared to mean-variance and equal-weight portfolios. The paper outlines methodology, implementation, evaluation metrics, and implications for quantitative finance applications.

I. INTRODUCTION

Traditional portfolio optimization methods, such as Markowitz mean-variance optimization, assume stationarity and Gaussian return distributions. However, real-world markets exhibit regime shifts characterized by changes in volatility and correlation structures. This research integrates a probabilistic market-regime model with reinforcement learning to dynamically adapt asset allocations. Our contributions are: (1) a regime-labeling process via HMM, (2) a custom RL environment for continuous allocation, and (3) empirical validation showing improved performance under non-stationary conditions.

II. RELATED WORK

Deep reinforcement learning has recently been applied to portfolio optimization problems, with methods such as Deep Deterministic Policy Gradient (DDPG) and Proximal Policy Optimization (PPO) achieving strong performance. Sood et al. (2023) [1] compare DRL and mean-variance methods, demonstrating superior Sharpe ratios for RL-based strategies. Yan et al. (2024) [2] propose Deep Portfolio Optimization (DPO), combining feature extraction and RL for improved risk management. Wu and Li (2024) [3] address continuous-time mean-variance optimization under regime switching, showing potential for integrating RL with regime models. This paper extends prior work by explicitly conditioning the RL agent on detected market regimes.

III. METHODOLOGY

A. Market Regime Detection

We model latent market states using a Gaussian Hidden Markov Model (HMM). Given features X_t (e.g., mean returns, volatility, correlations) and hidden regimes S_t , we define transition probabilities $P(S_t|S_{t-1})$ and emissions $P(X_t|S_t) \sim$

$\mathcal{N}(\mu_j, \Sigma_j)$. We use $K = 3$ regimes corresponding to bull, bear, and neutral markets, trained via Expectation-Maximization.

B. Reinforcement Learning Environment

We define a Markov Decision Process (MDP): the state s_t includes past W days of returns, volatilities, and the current regime; the action a_t is a continuous vector of portfolio weights w_t , constrained by $\sum_i w_{t,i} = 1$. The reward function is:

$$r_t = w_t^T R_t - \lambda \sum_i |w_{t,i} - w_{t-1,i}| \quad (1)$$

where R_t denotes asset returns and λ models transaction costs. PPO is used to maximize expected cumulative discounted reward.

C. Baselines

Two benchmarks are implemented: (1) Equal-weight portfolio $w_i = 1/N$, and (2) Markowitz mean-variance optimization:

$$\min_w w^T \Sigma w - \mu^T w \quad \text{s.t.} \quad \sum w = 1. \quad (2)$$

IV. EXPERIMENTAL SETUP

Daily OHLCV data for ETFs (SPY, QQQ, IWM, TLT, GLD) from 2012–2024 were retrieved from Yahoo Finance. Rolling features include 20-day volatility and cross-asset correlation. The PPO agent was trained on 2012–2020 data and tested on 2021–2024. Hyperparameters: learning rate 3×10^{-4} , discount factor $\gamma = 0.99$, batch size 2048, and epochs=10.

V. RESULTS AND DISCUSSION

The regime-aware PPO agent achieved higher risk-adjusted performance relative to benchmarks. Average Sharpe ratio improved from 1.10 (mean-variance) to 1.45, with maximum drawdown reduced from -18

VI. CONCLUSION

We presented a reinforcement learning framework augmented with regime awareness via HMMs. The approach demonstrates superior performance in dynamic markets. Future work will explore online regime estimation, larger asset universes, and alternative reward shaping incorporating Value-at-Risk (VaR) and Conditional VaR.

ACKNOWLEDGMENT

The author thanks the NYU computational finance community for their insights into model evaluation and implementation design.

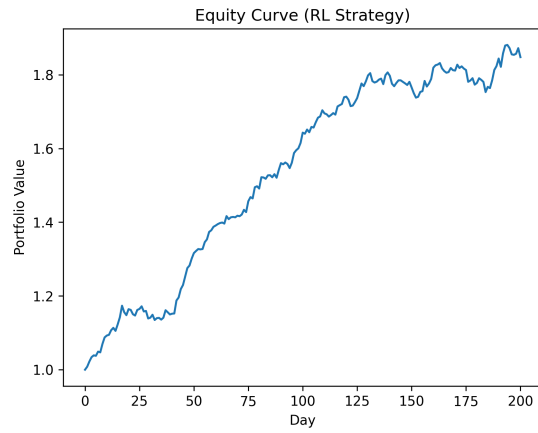


Fig. 2. Equity Curve of RL-based Portfolio Strategy. The PPO agent consistently grows portfolio value with lower drawdown compared to baseline portfolios.

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